12.7 Who Meets Whom in a Knockout

WORKSHEET

The question

STEP II Specimen paper, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q16

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself How many matches are played in a knockout tournament with 2^n players? (Each match eliminates one player.)

STAGE 2 The key symmetry

Ask yourself The draw in each round is random and every match is a fair coin toss. Does the process treat any pair of players differently from any other?

STAGE 4 At any stage

Ask yourself X and Y can meet at most once. How does that let you add probabilities over matches?

What's really going on

Expected counts

The argument in Stage 4 is really an expectation calculation: the number of times X and Y meet is a sum of indicators, one per match, and its expectation (which equals the probability of meeting, since it is 0 or 1) is the sum of the indicators' expectations. This is linearity of expectation, used in Section 9.11 and Section 12.4. Its great advantage is that it ignores all the complicated dependence between the rounds.

Seeded tournaments

Real tournaments are *seeded*: the strongest players are placed so that they cannot meet until the late rounds. And real matches are not 50–50. Both destroy the symmetry, and calculating the chance that two given players meet becomes a genuine computation. Statisticians who model sport use exactly such calculations, with each match's probability estimated from players' ratings, to forecast tournaments. The symmetric case is the baseline against which the real effects are measured.

Try this next

1. **Round by round.** Show directly that the probability that X and Y meet in round k is $\frac{2^{n-k}}{\binom{2^n}{2}}$, and check that these add up to $\frac{1}{2^{n-1}}$.
2. **A fixed bracket.** If instead the bracket is fixed in advance (as in the figure) but all players are placed in it at random, does the answer change?
3. **A strong player.** Suppose player X wins every match. What is now the probability that X meets Y?
4. **A league.** In a league of $2n$ teams where every pair meets once, how many matches are there? What is the analogous "probability of meeting"?

Further reading

- **Frederick Mosteller, *Fifty Challenging Problems in Probability*** (Dover), which includes tournament problems.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican), on modelling and forecasting.
- **Joseph Blitzstein & Jessica Hwang, *Introduction to Probability*** (CRC Press; free online), on indicators and symmetry.

The full guided solution is in Section 12.7:
https://beyondstep.keithmcnulty.org/ch12/knockout.html